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Institute of Digital Technology

HOSPITAL MANAGEMENT SYSTEM

Teacher : Ran chily

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1. Introduction

A Hospital Management System (HMS) is a web-based application designed to manage hospital operations efficiently. It helps in handling patient records, doctor schedules, appointments, and basic hospital services in a digital and organized way.

This system reduces paperwork, saves time, and improves communication between patients, doctors, and staff.

2. Objectives

- Manage patient and doctor information
- Schedule and track appointments
- Reduce paperwork and manual processes
- Improve accuracy and data access
- Manage billing and financial records

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3. Scope and Limitations

Scope

- Patient Management
- Doctor Management
- Appointment Booking
- Staff Management
- Billing & Invoicing
- Dashboard
- User Experience

Limitations

- No real database
- No real-time updates
- No external API or payment integration
- Frontend only – no backend

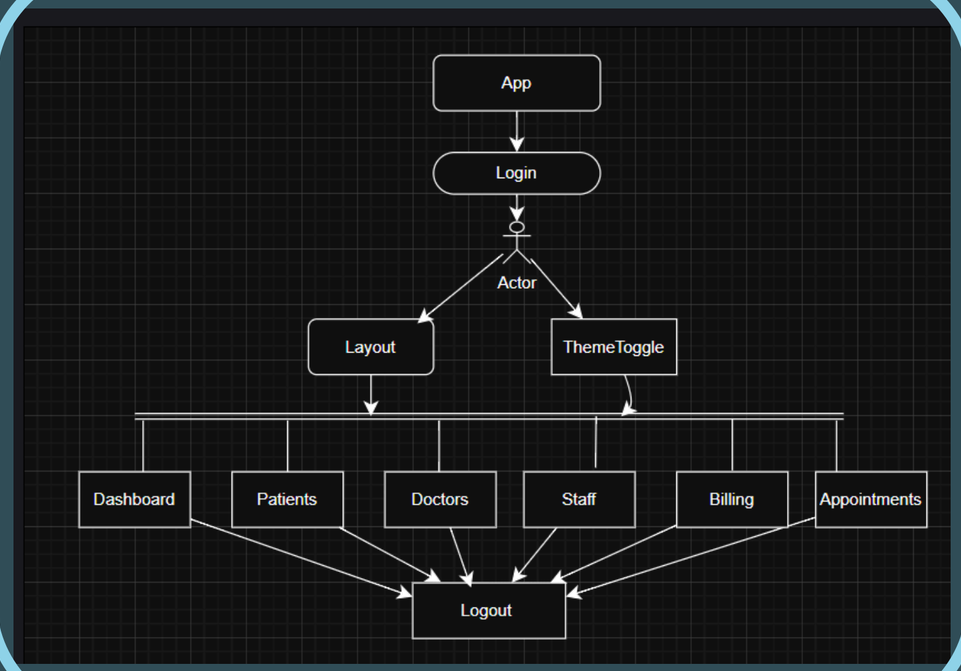
4. Methodology

a. Tools & Technologies

- React , React Router DOM , Custom CSS , Material Symbols ,Git ,VScode ,Vite ...

b. Use Case Diagram

- **Actor:** Hospital Administrator
- **Actions:** Login, Dashboard, Manage Patients, Doctors, Appointments, Staff, Billing, Theme Toggle, Logout



4. Methodology

Algorithm – Add Patient

1. Click "Register Patient" → modal opens
2. Fill form → each keystroke updates formData
3. Click "Save" button
4. Create object → new ID, status = Active, date = today
5. Prepend to array → newest record at top
6. Close modal → reset form
7. Table re-renders → new patient visible

Same pattern for Doctors, Appointments, Staff, Billing.

5. Results

- Successfully developed a functional Hospital Management System
- Users can add, edit, and manage patients, doctors, and appointments
- The system performs accurate data processing and updates
- The system supports efficient organization and management of hospital operations

6. Conclusion

- This project provides a reliable Hospital Management System
- It improves data accuracy and workflow efficiency
- It shows how software can replace manual processes and improve hospital operations

Demo

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Thank You